

Report 41 - FrancisTurbine Model

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Abstract

This report documents the FrancisTurbine model family in OpenHPLjl. Catalogue status: **prototype**.

1 Updated Engineering Context

Mechanistic Francis reaction-turbine family. It belongs to the Turbines domain.

2 Generalized Concept Model

The model preserves the domain connector interface while allowing internal fidelity to evolve independently.

3 Alternative Models Available in Literature

This family sits on the common fidelity ladder: boundary/algebraic, lumped dynamic, nonlinear physics-based, and distributed/high-order.

4 Governing Mathematics

$$P_m = \rho g Q H \eta, \quad T_m = P_m / \omega$$

The production model may add states, constitutive relations, lookup maps, or spatial discretization.

5 OpenHPL / Modelica Interpretation

Report 9 maps this family to the closest OpenHPL, Modelica Standard Library, or OpenIPSL analogue.

6 Julia / ModelingToolkit Implementation

Source: `src/Turbines/FrancisTurbine.jl`. The file is either an implemented component or an explicit Julia source specification.

7 Component Connections and Assembly

The model connects through the stable physical connector for its domain and should not expose internal fidelity choices to neighbouring components.

8 Numerical Experiment / Validation

Promotion requires an equation-level test, component experiment, limiting-case comparison, structural check, and connected-system test.

9 Expected Outputs

Expected outputs are the physical states, connector variables, and conservation or residual quantities needed to verify this family.

10 Engineering Interpretation

Build after lookup-model validation.

Conclusion

FrancisTurbine now has an explicit source/report pair. Promotion to baseline requires validated equations, parameters, tests, and reference behaviour.